XIAOMENG XU

https://xxm19.github.io/ | xuxm@stanford.edu

EDUCATION

Stanford University 09/2023-06/2027 (expected)
PhD Candidate in Electrical Engineering, Advisor: Prof. Shuran Song

Stanford University 09/2023-06/2025
Masters of Science in Electrical Engineering

Tsinghua University 08/2019-06/2023
Bachelor of Engineering in Automation (GPA 3.96/4.0), Bachelor of Arts in Industrial Design

RESEARCH INTERESTS

My research explores how machine learning expands the space of robots we can build and control, by co-designing robot sensing, morphology, and learning systems. This spans whole-body dexterity from distributed sensing, generative morphology-control co-design, and cross-embodiment learning from human demonstrations.

AWARDS AND HONORS

Fellowship
Stanford Interdisciplinary Graduate Fellowship (36 students across all departments) 05/2025-now

Paper Awards
Best Paper Award, Bimanual Robot Learning Workshop, IROS 2026 (top 1 of 11 accepted papers, $1000 prize) 09/2026
Best Paper Award, ScaleInfra Workshop, IROS 2026 (top 1 of 23 accepted papers) 09/2026
Best Paper Award, Human-to-Robot Workshop, CoRL 2025 (top 1 of 26 accepted papers) 09/2025
Best *Machine Learning* Paper Award, Morphology-Aware Policy and Design Learning Workshop, CoRL 2024 (top 1 of 12 accepted papers) 11/2024

Scholarships and Academic Honors
Outstanding Graduate, Tsinghua University (top 2% of graduates) 06/2023
Comprehensive Excellence Award, Tsinghua University (top 5%) 10/2022
National Scholarship, Government of China (highest national award, top 0.1%) 10/2021
129 Scholarship, Tsinghua University (highest sophomore award, top 1%) 10/2020

Travel Grants
NeurIPS Scholar Award 2025
CVPR Travel Grant 2023

PREPRINTS AND MANUSCRIPTS IN PREPARATION

1. Sha Yi, **Xiaomeng Xu**, Xiaolong Wang, Shuran Song, Mike Tolley, *Machine Learning for Robot Design.*
   Nature (review proposal accepted; manuscript in preparation)
2. **Xiaomeng Xu**, Han Zhang, Zhian Ruan, Xiaofeng Guo, Jinkun Cao, Ruihan Yang, Carmelo Sferrazza, Shuran Song†, Guanya Shi†, *PanTact: Teaching Whole-Body Dexterity with a Wearable Contact Interface.*
   arXiv preprint (2026)
   [Website] [Paper] [Code]

PUBLICATIONS

1. **Xiaomeng Xu***, Yifan Hou*, Chendong Xin, Zeyi Liu, Shuran Song, *Compliant Residual DAgger: Improving Real-World Contact-Rich Manipulation with Human Corrections.*
   International Journal of Robotics Research (**IJRR 2026**)
   Neural Information Processing Systems (**NeurIPS 2025**)
   Human-to-Robot Workshop, CoRL 2025 — **Best Paper Award**
   [Website] [Paper] [Code]
2. Yimeng Qin*, **Xiaomeng Xu***, William Heap, Aditi Oak, Shuran Song†, Allison Okamura†, *PanoVine: Whole-Body Visuomotor Control for Soft Growing Vine Robot.*
   Conference on Robot Learning (**CoRL 2026**)
   [Website] [Paper] [Code]
3. Dian Wang*, Jisang Park*, **Xiaomeng Xu**, Han Zhang, Shuran Song†, Jeannette Bohg†, *Mixture of Frames Policy: Multi-Frame Action Denoising for Bimanual Mobile Manipulation.*
   Conference on Robot Learning (**CoRL 2026**)
   [Website] [Paper] [Code]
4. **Xiaomeng Xu**, Jisang Park, Han Zhang, Eric Cousineau, Aditya Bhat, Jose Barreiros, Dian Wang, Jeannette Bohg, Shuran Song, *HoMMI: Learning Whole-Body Mobile Manipulation from Human Demonstrations.*
   Robotics: Science and Systems (**RSS 2026**)
   Bimanual Robot Learning Workshop, IROS 2026 — **Best Paper Award**
   ScaleInfra Workshop, IROS 2026 — **Best Paper Award**
   [Website] [Paper] [Video] [Code]
5. Ryan Punamiya, Simar Kareer, Zeyi Liu, Josh Citron, Ri-Zhao Qiu, Xiongyi Cai, Alexey Gavryushin, Jiaqi Chen, Davide Liconti, Lawrence Y. Zhu, Patcharapong Aphiwetsa, Baoyu Li, Aniketh Cheluva, Pranav Kuppili, Yangcen Liu, Dhruv Patel, Aidan Gao, Hye-Young Chung, Ryan Co, Renee Zbizika, Jeff Liu, **Xiaomeng Xu**, Haoyu Xiong, Geng Chen, Sebastiano Oliani, Chenyu Yang, Xi Wang, James Fort, Richard Newcombe, Josh Gao, Jason Chong, Garrett Matsuda, Aseem Doriwala, Marc Pollefeys, Robert Katzschmann, Xiaolong Wang, Shuran Song, Judy Hoffman, Danfei Xu, *EgoVerse: An Egocentric Human Dataset for Robot Learning from Around the World.*
   Robotics: Science and Systems (**RSS 2026**)
   [Website] [Paper] [Code]
6. Hojung Choi*, Yifan Hou*, Chuer Pan, Seongheon Hong, Austin Patel, **Xiaomeng Xu**, Mark Cutkosky, Shuran Song, *In-the-Wild Compliant Manipulation with UMI-FT.*
   IEEE International Conference on Robotics and Automation (**ICRA 2026**)
   [Website] [Paper] [Hardware & Code]
7. **Xiaomeng Xu**, Dominik Bauer, Shuran Song, *RoboPanoptes: The All-Seeing Robot with Whole-body Dexterity.*
   Robotics: Science and Systems (**RSS 2025**)
   [Website] [Paper] [Video] [Hardware & Code]
8. Haoyu Xiong, **Xiaomeng Xu**, Jimmy Wu, Yifan Hou, Jeannette Bohg, Shuran Song, *Vision in Action: Learning Active Perception from Human Demonstrations.*
   Conference on Robot Learning (**CoRL 2025**)
   [Website] [Paper] [Video] [Hardware & Code]
9. **Xiaomeng Xu**, Huy Ha, Shuran Song, *Dynamics-Guided Diffusion Model for Sensor-less Robot Manipulator Design.*
   Conference on Robot Learning (**CoRL 2024**)
   Morphology-Aware Policy and Design Learning Workshop, CoRL 2024 — **Best Machine Learning Paper Award**
   [Website] [Paper] [Video] [Code]

10. **Xiaomeng Xu***, Yanchao Yang*, Kaichun Mo, Boxiao Pan, Li Yi, Leonidas Guibas, *JacobiNeRF: NeRF Shaping with Mutual Information Gradients.*
Conference on Computer Vision and Pattern Recognition (**CVPR 2023**)
[Website] [Paper] [Video] [Code]

11. Yun Liu*, **Xiaomeng Xu***, Weihang Chen, Haocheng Yuan, He Wang, Jing Xu, Rui Chen, Li Yi, *Enhancing Generalizable 6D Pose Tracking of an In-Hand Object with Tactile Sensing.*
Robotics and Automation Letters (**RA-L 2023**), presented at ICRA 2024
[Paper] [Video] [Code]

12. Xueyi Liu, **Xiaomeng Xu**, Anyi Rao, Chuang Gan, Li Yi, *AutoGPart: Intermediate Supervision Search for Generalizable 3D Part Segmentation.*
Conference on Computer Vision and Pattern Recognition (**CVPR 2022**)
[Website] [Paper] [Video] [Code]

13. Guanhong Liu, Tianyu Yu, Zhihao Yao, Haiqing Xu, Yunyi Zhang, Xuhai Xu, **Xiaomeng Xu**, Mingyue Gao, Qirui Sun, Tingliang Zhang, Haipeng Mi, *ViviPaint: Creating Dynamic Painting with a Thermochromic Toolkit.*
Multimodal Technologies and Interaction (**MTI 2022**)
[Website] [Paper]

* equal contribution, † equal advising

INVITED TALKS

Dexterity Through Hardware and Learning Co-Design
CHARM Lab, Stanford University — 10/2026

Robots That See and Act with Their Whole Body
Sundae Robotics — 10/2026
BeNeRL Seminar — 10/2026
Physical AI After Hours, IROS 2026 — 09/2026
Graphics Cafe, Stanford University — 05/2026

From Human Demonstrations to Human Corrections
RCHI Lab, Carnegie Mellon University — 04/2026

HoMMI: Learning Whole-Body Mobile Manipulation from Human Demonstrations
LeCAR Lab, Carnegie Mellon University — 04/2026

Learning Whole-Body Manipulation via Hardware-Policy Co-design
Shanghai Jiao Tong University — 03/2026

Embodiment-Aware Robot Learning: Co-designing Robot Morphology, Sensing, and Control
The Helping Hands Lab, Northeastern University — 08/2025

INTERNSHIP EXPERIENCE

Frontier AI and Robotics Team @ Amazon — 06/2026-09/2026
Research Intern, Mentors: Prof. Guanya Shi, Prof. Carmelo Sferrazza — *San Francisco, CA*

· Learning humanoid whole-body loco-manipulation from human demonstrations

Large Behavior Models Team @ Toyota Research Institute — 06/2025-09/2025
Research Intern, Mentor: Eric Cousineau — *Cambridge, MA*

· Scaling whole-body mobile manipulation with egocentric human demonstrations

Stanford Geometric Computing Lab 06/2022-08/2022
Research Intern, Advisor: Prof. Leonidas Guibas *Stanford, CA*

- Chinese Undergraduate Visiting Research Program (UGVR).
- Shaping a NeRF to encode mutual correlations of a scene via aligning Jacobians.

SERVICES

Workshop Co-organizer
Diffusion for Robot Learning, RSS 2026 06/2026
Reinforcement Learning in the Era of Imitation Learning, ICRA 2026 06/2026
1st Workshop on Robot Hardware-Aware Intelligence, RSS 2025 06/2025

Associate Editor
IEEE International Conference on Robotics and Automation (ICRA), 2027

Reviewer
Conferences: RSS (2025), CoRL (2025–2026), ICRA (2024, 2026), IROS (2025), NeurIPS (2024–2026), ICLR (2025–2027), ICML (2025), CVPR (2023), AAAI (2025)
Journals: IEEE Transactions on Robotics (2025–2026), Robotics and Automation Letters (2025–2026)

Teaching
Teaching Assistant, EE/CS227: Robot Perception, Stanford University Fall 2024–2025

Community and Outreach
Member, *Women in Robotics*, International professional network 08/2026–now
Member, *Women in Electrical Engineering*, Stanford University 10/2023–now
Member, *Women in Computer Science*, Stanford University 10/2023–now
Participant, *Engineering Students for DEI*, Stanford University 4/2025–now
Participant, *Women in Mechanical Engineering*, Stanford University 10/2023–now
Participant, *TRI Women & Allies*, Toyota Research Institute 06/2025-09/2025
Drop-in tutor for engineering drawing, CAD, programming, circuits, physics, and calculus, Tsinghua University 10/2020-10/2022